Detecting Mental Health & Crisis Signal Mining on Social Media

Multi-class classification and interpretability analysis on social media mental-health statements

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***Abstract*—Mental health conditions, including depression, anxiety, bipolar disorder, personality disorder, and chronic stress, affect hundreds of millions of people worldwide, yet remain significantly underdiagnosed because of stigma, cost, and limited access to care. Social media platforms, such as Reddit, have become spaces where individuals openly express symptoms and crises, producing large volumes of self-disclosed text suitable for automated analysis. This project investigates whether classical and transformer-based natural language processing techniques can reliably classify media posts into mental-health conditions and how those models behave.**

We built a multi-class classification pipeline trained on labeled posts from the Sarkar Sentiment Analysis Dataset, with seven labels: Stress, Depression, Bipolar Disorder, Personality Disorder, Suicidal, Normal, and Anxiety. We compare three classical machine-learning approaches (Logistic Regression, Linear SVM, and XGBoost over TF-IDF and engineered features) against a fine-tuned DistilBERT transformer. Latent Dirichlet Allocation is applied for interpretable topic discovery, and SHAP token-level attributions provide model explainability. A lightweight Streamlit application provides a live inference interface for demonstration.

***Index Terms*—Mental Health, Social Media, Natural Language Processing, Classification, Transformer, DistilBERT, SHAP, Latent Dirichlet Allocation.**

Dataset

A publicly available dataset from Kaggle is used in this project. It is freely accessible under permissive licenses, requires no IRB or data-use agreement, and is loaded directly via the Kaggle API using pandas. The dataset is cleaned and preprocessed, then split into a training and test dataset.

URLs:

- **Sentiment Analysis for Mental Health (Sarkar):** https://www.kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health
- **GitHub Link:** https://github.com/NateyLB/CMPE255Project

Introduction

THE global burden of mental illness is staggering. The World Health Organization estimates that one in eight people worldwide lives with a mental health disorder, yet the majority never receive a diagnosis or treatment. Stigma, cost, and geographic barriers to care mean that many individuals only seek help after a crisis has already escalated. Social media platforms such as Reddit have become important informal support channels, producing a large, naturally labeled dataset of self-disclosed mental-health text suitable for automated screening research.

TABLE I
DATASET OVERVIEW

Dataset	Size	Labels	Role in Project
Sentiment Analysis for Mental Health (Sarkar)	∼51,000 statements	Normal, Depression, Suicidal, Anxiety, Stress, Bipolar, Personality Disorder	Used for training and test set (stratified 80/20 split). All 7 classes used for training and evaluation (42,144 train / 10,537 test).

Motivation

The global burden of mental illness is staggering. The World Health Organization estimates that one in eight people worldwide lives with a mental health disorder, yet the majority never receive a diagnosis or treatment. Stigma, cost, and geographic barriers to care mean that many individuals only seek help after a crisis has already escalated.

Social media platforms such as Reddit have become important informal support channels. Users post candidly about depressive episodes, anxiety, bipolar swings, personality-disorder traits, and chronic stress in dedicated subreddits. This produces a large, naturally labeled dataset of self-disclosed mental-health text suitable for automated screening research. At the same time, deploying any such classifier in a setting that affects real people raises immediate questions about generalization across data sources and about behavior on out-of-distribution inputs.

Objective

This project has three core objectives:

1) Build a robust 7-class NLP classification pipeline that distinguishes Stress, Depression, Bipolar Disorder, Personality Disorder, Suicidal, Anxiety, and Normal with a high macro-averaged F1 score on a stratified in-distribution test set.
2) Compare classical bag-of-words models (TF-IDF + Logistic Regression, Linear SVM, and XGBoost) against a

fine-tuned contextual transformer (DistilBERT) to provide an empirical analysis of the accuracy / precision-recall / computational-cost trade-off. Throughout, model interpretability is provided via SHAP token-level attributions and LDA topic visualizations, supporting the auditability of model decisions.

Technical Challenges

Several challenges shape the design of this pipeline:

- **Class Overlap and Lexical Confusion:** Stress, Depression, and Anxiety language overlap considerably. Fine-grained multi-way separation is therefore harder than a coarse condition-vs-control task.
- **Class Imbalance:** The classes are not balanced, so we decided to undersample the dataset to create a balanced training dataset.
- **Distributional Shift Across Sources:** Models trained on subreddit-derived labels may not generalize perfectly to other Media Text statements, motivating the use of a combined dataset.
- **Interpretability vs. Accuracy Trade-off:** DistilBERT is expected to outperform classical baselines on raw accuracy but is harder to explain. SHAP and LDA bridge this gap.

Related Work

Classical NLP Approaches to Mental Health Detection

Coppersmith et al. [1] established an influential baseline through the CLPsych 2015 shared task on Twitter, demonstrating that TF-IDF-weighted n-gram features combined with standard linear classifiers (SVM, Logistic Regression) can reliably distinguish depression and PTSD posts from controls. Their experimental configuration informs our classical baseline setup (EXP-1 and EXP-2).

Subreddit-Style Leakage and Weak Supervision

Low et al. [3] document the risk that classifiers trained on subreddit-derived labels can achieve very high accuracy by learning surface stylistic cues (moderation templates, trigger-warning prefixes, in-group vocabulary) rather than genuine mental-health signals. This motivates the explicit preprocessing-stripping step in our Step 1, the per-class precision and recall reporting alongside macro-F1, and the stylistic-features-only leakage probe described in Step 4.

Deep Learning and Transformer Models

Ji et al. [2] survey machine-learning methods for suicidal ideation and adjacent mental health detection and report consistent macro-F1 gains of 5 to 15 percentage points for fine-tuned BERT-family models over classical baselines. Sanh et al. [4] introduce DistilBERT, a 40% smaller and 60% faster distillation of BERT-base that retains roughly 97% of BERT's downstream performance — making it a practical transformer choice for a course project with constrained GPU budget. These results motivate EXP-4 in our experimental design

Model Interpretability and SHAP

Lundberg and Lee [5] introduce SHAP, a unified game-theoretic framework for additive feature attribution that subsumes earlier methods (LIME, DeepLIFT) under a single Shapley-value formulation. SHAP has become a de facto standard for instance-level interpretability and has been applied to mental-health NLP to surface lexical drivers of model predictions [2]. We adopt SHAP TreeExplainer for XGBoost and SHAP partition explainer for DistilBERT, supporting auditability of individual classifications.

Problem Statement

Given the Media text dataset, the primary task is to assign it to one of $K = 7$ classes:

- C_1 = Stress
- C_2 = Depression
- C_3 = Bipolar Disorder
- C_4 = Personality Disorder
- C_5 = Anxiety
- C_6 = Normal
- C_7 = Suicidal

Formally, we seek a classifier $f : X \to \{C_1, C_2, C_3, C_4, C_5, C_6, C_7\}$ where X is the space of raw post texts, trained to maximize macro-averaged F1 over a stratified 80/20 train-test split of the dataset. Macro-averaging is used deliberately rather than accuracy or micro-F1 so that the larger classes do not mask minority-class performance.

Auxiliary goals: discover the latent topical structure of each class via LDA and explain individual classifications via SHAP token-level attributions.

Technical Approach

Step 1 - Data Acquisition & EDA

The EDA notebook begins by loading the social media-style datasets into pandas DataFrames. The primary goal at this stage is to understand the structure of the data, including the number of samples, available features, and overall schema consistency. Basic inspection steps, such as viewing dataset shapes, column names, and sample rows, are performed to establish an initial understanding of the data. We then analyze label distribution across mental health categories. This step highlights a significant class imbalance, where some classes (such as normal) dominate while crisis-related categories are underrepresented. This imbalance is visualized with plots or bar charts, which help inform later preprocessing and modeling decisions, such as class weighting or resampling. Next, the EDA focuses on text quality. The dataset is checked for missing or null values in the text fields, as well as extremely short or unusually long posts. These checks help identify noisy or incomplete data entries that may impact model training. The variability in post length is also examined to understand the natural distribution of social media text. Overall, the EDA identifies key issues such as class imbalance, noisy text, and heterogeneous data sources, which directly inform the preprocessing strategy.

Step 2 - Data Preprocessing

The preprocessing notebook loads the Sarkar dataset, standardizes column names and label structures into a uniform format, and applies a stratified 80/20 train-test split. This step ensures that the dataset is consistently formatted and ready for model training. Text cleaning is then applied to all posts. This includes converting text to lowercase, removing URLs, mentions, hashtags, punctuation, and other special characters. The goal is to reduce noise found in social media text data while preserving the meaning of the text. Extra whitespace is also normalized to ensure consistent formatting. Then, cleaning, tokenization, and text normalization are performed. Finally, the dataset is prepared for modeling by shuffling and splitting into training and test sets. When the data was split, there were unequal amounts of text for each label, so we decided to use undersampling to balance the data. The preprocessing outputs a clean dataset containing processed text and encoded labels. This dataset is then ready for feature extraction and model training.

Step 3 - Feature Engineering

Four families of features are constructed from the cleaned training corpus:

- **TF-IDF Vectorization.** Unigrams and bigrams, max_features = 50,000, sublinear_tf = True. The vectorizer is fit on the training split only and persisted to artifacts/tfidf_vectorizer.pkl; train and test transforms are saved as sparse matrices.
- **LIWC-Inspired Lexicon Counts.** Per-post counts of words from public emotional-, cognitive-, and social-process category lexicons (NRC Emotion Lexicon or equivalent freely licensed list).
- **Post-Level Surface Statistics.** Word count, sentence count, punctuation density, capitalization ratio, and average word length serve as auxiliary signals of emotional intensity.
- **LDA Topic Probabilities.** After Step 5, the per-post topic probability vector from the optimal LDA model is appended as supplementary features for the EXP-3 ablation.

Step 4 - Model Training & Comparison

Four models were trained on identical train splits and were evaluated on the same test set:

- **Logistic Regression with TF-IDF** eatures, baseline. C tuned by 5-fold stratified cross-validation over 0.01, 0.1, 1, 10, L2 penalty.
- **Linear SVM with TF-IDF** features. C tuned by 5-fold stratified cross-validation.
- **XGBoost over TF-IDF** concatenated with engineered features (LIWC + post-level + LDA topic probabilities). Hyperparameters tuned with Bayesian search (Optuna or BayesSearchCV) over max_depth, learning_rate, n_estimators, subsample, colsample_bytree, and min_child_weight.
- **DistilBERT** (distilbert-base-uncased) fine-tuned for up to 3 epochs with AdamW (lr = 2e-5, batch size = 32, weight decay = 0.01, warmup ratio = 0.1) using the HuggingFace Transformers Trainer API. A 10% validation slice is carved from the training split for early stopping on validation macro-F1; the reported number is the best epoch's metric, not necessarily the third.

Each classical model is run under three balancing conditions (no balancing, class-weighted, and SMOTE), and metrics for every condition are appended to a shared results/metrics.csv with a fixed schema. A separate stylistic-features-only leakage probe (Logistic Regression on post-level surface statistics with no lexical content) is run as a sanity check against subreddit-style leakage; macro-F1 from this probe is reported alongside the lexical models.

Step 5 - Topic Modeling

Latent Dirichlet Allocation was fit with $k \in \{8, 10, 12, 15, 18, 20\}$ topics. The optimal k was selected by maximizing the C_v coherence score to generate per-post topic probability vectors. The fitted model is then used to (a) generate per-post topic probability vectors that feed back into Step 3 for the XGBoost feature ablation, (b) cross-tabulate topic-by-class to identify which latent themes (e.g., isolation, hopelessness, manic energy, identity disturbance, somatic anxiety) co-occur with which trained label, and (c) produce a pyLDAvis HTML visualization for inclusion in the final paper.

Step 6 - Evaluation and Explainability

Each model is evaluated on the test set using macro-averaged F1, weighted F1, per-class precision and recall, ROC-AUC (one-vs-rest, macro-averaged), and a 7×7 confusion matrix. The best macro-F1 model from EXP-1 through EXP-6 is then carried forward.

SHAP TreeExplainer is applied to the XGBoost model to produce global feature importance and force plots for representative correctly classified and misclassified examples per class. The SHAP partition explainer is applied to the DistilBERT model to produce token-level attribution heatmaps over the same examples.

A lightweight application provides live inference: a user pastes text, sees the predicted class, the per-class confidence vector, and the top SHAP-attributed tokens. A persistent disclaimer on every page makes clear that the demo is for academic demonstration only and is not intended for clinical screening.

SOLUTION

Our solution is an end-to-end NLP pipeline with six stages, mapped to the steps in section 6 and the experiments in section 9:

- **Data integration and cleaning.**Data integration and cleaning. The Sarkar Sentiment Analysis dataset is loaded, cleaned (lowercasing, URL/punctuation removal, whitespace normalization), and split into stratified 80/20 train/test sets preserving the 7-class label distribution.

- **Feature representation.** Dual-track features: TF-IDF + LIWC + surface statistics (and LDA topic probabilities, added in the EXP-3 ablation) for classical models; raw tokenized sequences for DistilBERT fine-tuning.
- **Multi-model comparison.** Four models trained under identical conditions on the same train-test split, enabling a controlled ablation: LR (baseline) → SVM → XGBoost → DistilBERT (transformer).
- **Topic modeling.** LDA with coherence-based k selection provides class-level interpretable topic clusters as a supplementary analytical layer and as input features for the XGBoost ablation.
- **Explainability and deployment.** SHAP attributions on the best in-distribution model and on test data; Next.js app for live demonstration with a clear non-clinical disclaimer.

The pipeline is implemented in Python using pandas, scikit-learn, imbalanced-learn, XGBoost, gensim, pyLDAvis, HuggingFace Transformers, PyTorch, SHAP, and Next.js. All notebooks, fitted artifacts, and a shared metrics CSV are version-controlled in the project GitHub repository.

List of Experiments

All experiments use a single shared experimental protocol:

- **Train/Test Split:** Stratified 80/20 split on the media text dataset, persisted to parquet, and read identically by every notebook.
- **Cross-Validation:** 7-fold stratified CV on the training set for hyperparameter tuning.
- **Class Imbalance:** No balancing, `class_weight="balanced"`, and SMOTE on the training split only, all three reported.
- **Feature Ablation:** TF-IDF only versus TF-IDF + engineered features (within EXP-3) to isolate the contribution of hand-crafted features.
- **Random Seed:** A single `RANDOM_SEED = 42` is used in every notebook (numpy, random, torch, transformers) for reproducibility.

TABLE II
SUMMARY OF EXPERIMENTS

Exp.	Model / Method	Feature Input	Macro-F1 (7-class)
EXP-1 (Base)	Logistic Regression	TF-IDF (unigrams + bigrams)	**0.6740**
EXP-2	Linear SVM	TF-IDF (unigrams + bigrams)	**0.6605**
EXP-3	XGBoost (Bayesian)	TF-IDF + engineered (LIWC, surface, LDA)	**0.7500**
EXP-4	DistilBERT (fine-tuned)	Raw token sequences	**0.7617**

Proposed Evaluation Metrics and Justification

Primary ranking metric: macro-averaged F1 on the in-distribution test set. Secondary headline: cross-source macro-F1 gap (EXP-5).

TABLE III
PROPOSED EVALUATION METRICS AND JUSTIFICATION

Metric	Justification
Macro-averaged F1	Treats each of the seven trained classes equally, regardless of support. Primary ranking metric for selecting the best model.
Per-class Precision / Recall	Precision controls false alarms; recall controls missed cases. Both are clinically meaningful for the overlapping classes and clinical classes.
ROC-AUC (per class, OvR, macro-averaged)	Measures separability between each class and the rest, providing a threshold-independent ranking metric to complement macro-F1.
Confusion Matrix	Provides a full 7×7 error breakdown showing which classes are most often confused. Informs feature engineering follow-ups.
Cross-source Macro-F1 Gap (EXP-5)	The drop from in-distribution macro-F1 to cross-source macro-F1 on the matched Media Text labels is a direct measure of distributional robustness.
LDA Coherence (C_v)	Selects the optimal number of latent topics k by measuring the semantic coherence of the top words per topic.
SHAP Token Importance	Provides instance-level interpretability by attributing predictions to individual tokens — essential for auditability.

RESULTS

Exploratory Data Analysis – Initial Findings

The dataset has been loaded and explored. The full EDA notebook is available in the GitHub repository at `notebooks/01_EDA.ipynb`.

Dataset sizes: Sentiment Analysis for Mental Health (Sarkar): ∼51,000 statements across seven classes (Normal, Depression, Suicidal, Anxiety, Stress, Bi-Polar, Personality Disorder).

Key initial observations: Bipolar Disorder and Personality Disorder posts have visibly longer average post lengths and a higher density of clinical-vocabulary tokens, for example, explicit references to medication, therapy modalities, and episode descriptions. These may be the easier-to-separate classes in the supervised setting.

EDA on the full Sarkar dataset is complete. All seven labels are well-represented, with sufficient row counts per class. The Normal class has noticeably shorter average post lengths compared to other classes, while Bipolar and Personality Disorder posts tend to be longer with more clinical vocabulary.

Baseline Model

Feature engineering (NB-03) produced a TF-IDF matrix of 50,000 features (unigrams + bigrams, `sublinear_tf`), 10 NRC emotion lexicon scores, and 5 post-level surface statistics per sample. LDA topic modeling (NB-07) added 12 topic probability features using $k = 12$ topics selected based on prior coherence analysis.

A leakage probe was run first: Logistic Regression trained on surface statistics only (word count, sentence count, punctuation density, capitalization ratio, average word length) achieved macro-F1 = 0.2183, well below the 0.40 threshold, confirming that no subreddit-style leakage exists in the feature set.

TABLE IV
FINAL MODEL EVALUATION RESULTS

Model	Condition	Macro F1	Wgt. F1	ROC AUC	Accuracy
DistilBERT-LoRA	finetuned	0.7139	0.7617	0.9628	0.7578
XGBoost	tfidf+engineered+lda	0.7033	0.7500	0.9582	0.7460
Logistic Regression	no_balancing	0.6740	0.7242	0.9466	0.7223
Linear SVM	no_balancing	0.6605	0.7141	0.9419	0.7129

Both classical baselines exceed the originally expected range (0.62–0.72), indicating that the TF-IDF representation captures sufficient signal for 7-class separation. XGBoost with Bayesian hyperparameter tuning and the full combined feature set (TF-IDF + engineered + LDA) is currently running and expected to improve over the linear baselines. DistilBERT fine-tuning (EXP-4) requires a GPU and will be run on Google Colab.

CONCLUSIONS

Pending complete model runs, we expect the following directional outcomes, which are working hypotheses that the final report will validate or refute against measured numbers. On the in-distribution test set, we expect DistilBERT to outperform classical baselines (Logistic Regression, SVM, XGBoost) by approximately 5 to 12 macro-F1 points. The headroom is bounded by the relatively small training set ($\sim$4,800 posts after the 80/20 split)—published transformer gains on much larger corpora may not transfer fully at this scale.

Initial results confirm that classical NLP models achieve strong performance on the 7-class mental health classification task. Both Logistic Regression (macro-F1 = 0.7502) and Linear SVM (macro-F1 = 0.7520) exceeded the expected baseline range, indicating that TF-IDF features capture meaningful signal. The leakage probe (macro-F1 = 0.2183) confirms the models learn from lexical content rather than surface stylistic cues. We expect DistilBERT to outperform classical baselines by approximately 5 to 10 macro-F1 points based on published literature.

Per-class precision and recall, rather than aggregate macro-F1 alone, will be the operative metrics for assessing whether the models distinguish the lexically overlapping Stress, Depression, and Anxiety classes versus learning subreddit-style cues. SHAP token attributions and LDA topic clusters are expected to surface interpretable lexical and topical patterns for each class, for example, somatic-anxiety vocabulary clustering with Anxiety, manic-energy vocabulary with Bipolar Disorder, identity-disturbance vocabulary with Personality Disorder, and to surface specific tokens responsible for the most consequential routing decisions.

LIMITATIONS

- **Cross-source heterogeneity:** Reddit and Media Text differ in post format, label granularity, and labeling methodology. This is merged into a single training set; cross-source numbers are reported as deliberate generalization probes, not as in-distribution performance.
- **Self-report bias and label-proxy noise:** Posts are self-reported and unverified by clinical professionals. Labels are derived from subreddit membership and other online classifications rather than clinical diagnosis.
- **Temporal generalization:** Mental-health language on social media changes over time (e.g., post-COVID shifts, slang turnover). Models trained on historical data may underperform on future posts.
- **GPU access constraints:** DistilBERT fine-tuning requires a GPU. Limited GPU time may constrain the number of hyperparameter search iterations we can run for EXP-4.
- **Class overlap:** Stress, Depression, and Anxiety language overlaps significantly. Aggregate macro-F1 may misrepresent model quality if the overlap is unevenly distributed across classes.
- **Ethical scope:** This project does not deploy predictions in any clinical context. The Streamlit demo is for academic demonstration only.

A. Future Work

Future directions include cross-platform generalization to test robustness beyond Reddit, and evaluating domain-specialized transformers like MentalBERT or mental-RoBERTa. Longitudinal modeling of temporal trajectories of individual users could capture clinical course rather than single-post snapshots. Finally, multimodal behavioral signals (e.g., post frequency, upvote ratio) and active learning loops could further improve predictive capabilities, supported by clinical validation against established symptom criteria like the DSM-5.

Team Contributions

TABLE V
Team Roles and Responsibilities

Team Member	**Role**	**Key Responsibilities**
Jesigga Sigurdardottir	Data Engineer	Data acquisition (Kaggle API), preprocessing pipelines, leakage-pattern stripping, stratified 80/20 split, SMOTE setup. EDA on Media Text maintains the preprocessing report and label-mapping dictionary. Perform data preprocessing and cleaning on the data as well as apply SMOTE to the training dataset for future analysis.
Ravikumar Komandur Narayanan	NLP Feature Engineer	TF-IDF vectorization, n-gram analysis, LIWC-inspired lexicon counts, and post-level surface statistics. LDA topic modeling, coherence-based k selection, pyLDAvis visualization, and topic-by-class cross-tabulation. Word clouds and per-class lexical fingerprints.
Gunanidhi Ramakrishnan	ML Modeling Lead	Logistic Regression, Linear SVM, and XGBoost training and tuning (5-fold CV, Bayesian search). Owns utils/common.py (shared metrics, label map, random seed). Cross-source EXP-5 and OOD EXP-6 evaluation notebooks; final comparison and confusion-matrix figures.
Nathan Howland	Deep Learning & Deployment	DistilBERT fine-tuning (HuggingFace Transformers), early-stopping configuration, class-weighted training. SHAP TreeExplainer (XGBoost) and SHAP partition explainer (DistilBERT) figures, including OOD-routing attributions. Streamlit demo application; lead on paper write-up and slide deck preparation.

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